INCH POUND

MIL-R-47211A(MI)
21 September 1989
SUPERSEDING
MIL-R-47211(MI)
12 July 1974

MILITARY SPECIFICATION

RUBBER, SILICONE, ROOM TEMPERATURE CURING

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general purpose, room temperature curing, silicone rubber compounds (see 6.1).

1.2 Classification. Compounds shall be of the following types, as specified (see 6.2):

Type I	Two Component
Type II	One Component (Acetic Acid Evolving)
Type III	One Component (Methyl Alcohol Evolving)
Type IV	One Component

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 9320

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SPECIFICATIONS

MILITARY

MIL-S-8802	Sealing Compound, Temperature-Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-45662	Calibration Systems Requirements

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Standard Test Method for
ASTM D 150	A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials, Standard Test Methods for
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Standard Test Methods for
ASTM D 412	Rubber Properties in Tension, Standard Test Methods for
ASTM D 624	Rubber Property - Tear Resistance, Standard Test Method for

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ASTM D 792 Specific Gravity (Relative Density) and Density of Plastics by Displacement, Standard Test Methods for

ASTM D 2240 Rubber Property - Durometer Hardness, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1960 Race Street, Philadelphia, Pennsylvania 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Typical properties. The values listed in the following tables are typical and may vary due to thickness of application, amount of catalyst used, temperature, relative humidity, and length of cure.

3.2.1 Uncured properties. The uncured compound shall exhibit the properties shown in table I, as applicable.

TABLE I. Uncured properties.

Property	Type I	Type II	Type III	Type IV
a. Specific gravity	1.50 ± .03	1.05 ± .04	1.12 ± .05	1.05 ± .05
b. Nonvolatiles (percent, minimum)	98	92	96	NA

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3.2.2 Cured properties. The cured compound shall exhibit the properties shown in table II, as applicable.

TABLE II. Cured properties.

Property	Type I	Type II	Type III	Type IV
a. Hardness, Shore A	50-65	20 Max.	25 Min.	18 Min.
b. Tensile strength, (psi) minimum	350	175	550	250 Min. 500 Max.
c. Elongation (percent), minimum	100	300	600	300
d. Tear resistance (lb/in), minimum	60	50	80	NA
e. Linear shrinkage (percent), minimum	1.0	1.0	NA	NA

3.2.3 High temperature aging properties. The cured compound shall exhibit the properties shown in table III, as applicable.

TABLE III. High temperature aging properties.

Property	Type I	Type II	Type III	Type IV
a. Hardness, Shore A				
24 hours at 315.5° Celsius (C) (600° Fahrenheit [F])	NA	30 ± 10	NA	NA
40 hours at 315.5° C	65 ± 10	NA	NA	NA
72 hours at 315.5° C (600° F)	NA	NA	60 Max.	NA

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TABLE III. High temperature aging properties (Continued).

Property	Type I	Type II	Type III	Type IV
b. Tensile strength (psi), minimum				
24 hours at 315.5° C (600° F)	NA	300	NA	NA
40 hours at 315.5° C (600° F)	600	NA	NA	NA
72 hours at 315.5° C (600° F)	NA	NA	200	NA
168 hours at 315.5° C (600° F)	NA	420	NA	NA
c. Elongation (percent), minimum				
24 hours at 315.5° C (600° F)	NA	300	NA	NA
72 hours at 315.5° C (600° F)	NA	NA	100% Min.	NA
168 hours at 315.5° C (600° F)	NA	300	NA	NA

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3.2.4 Electrical properties. The cured compound shall exhibit the properties shown in table IV, as applicable.

TABLE IV. Electrical properties.

Property	Type I	Type II	Type III	Type IV
a. Dielectric strength (volts/mil), maximum				
0.075 inch thick	300	300	450 Min.	400 Min.
b. Dielectric constant, maximum				
60 Hertz (Hz)	4.0	2.8	NA	NA
10^6 Hz	5.0	5.0	NA	NA
100 Hz	NA	NA	4.0	3.0
100 KHz	NA	NA	4.0	3.0
c. Dissipation factor, maximum				
60 Hz	0.020	0.0026	NA	NA
10^6 Hz	0.003	0.0026	NA	NA
100 Hz	NA	NA	0.003	.01
100 KHz	NA	NA	0.003	NA
d. Volume resistivity (ohm/cm^3), maximum	1.3×10^{14}	3×10^{15}	1×10^{13} Min.	1×10^{13} Min.

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3.3 Workmanship. The compound shall contain no contaminants and shall be packaged in containers which will preclude the entrance of contaminants, moisture, and air.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method documents and paragraphs 4.4 and 4.5 in this specification.

4.4 First article inspection. A first article sample consisting of 10 ± 2 pounds of types I, II, III and IV shall be provided. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production. The first article sample tests shall be as specified in table V.

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TABLE V. First article sample inspection.

Examination or Test	Requirement Paragraph	Method Paragraph
Specific gravity	3.2.1a	4.8.2.1
Nonvolatiles	3.2.1b	4.8.2.2
Hardness, Shore A	3.2.2a	4.8.3.1
Tensile strength	3.2.2b	4.8.3.2
Elongation	3.2.2c	4.8.3.3
Tear resistance	3.2.2d	4.8.3.4
Linear Shrinkage	3.2.2e	4.8.3.5
High temperature aging	3.2.3	4.8.5
Electrical properties	3.2.4	4.8.4
Workmanship	3.3	4.8.1

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified in table VI.

TABLE VI. Quality conformance inspection.

Examination or Test	Requirement Paragraph	Method Paragraph
Specific gravity	3.2.1a	4.8.2.1
Tensile strength	3.2.2b	4.8.3.2
Elongation	3.2.2c	4.8.3.3
Product marking	5.3	5.3
Workmanship	3.3	4.8.1
Packaging	Section 5	4.8

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4.5.1 Inspection lot. An inspection lot for quality conformance inspection shall consist of material manufactured under uniform conditions and offered for acceptance at one time.

4.5.2 Quality conformance sample. Quality conformance inspection shall be performed on not less than 5 ± 1 pounds taken from each lot of compound. Failure of the sample to meet the quality conformance requirements shall be cause for lot rejection.

4.6 Inspection equipment and facilities. The manufacturer shall insure that inspection equipment and facilities are of sufficient quality and quantity to permit performance of required inspections.

4.6.1 Inspection equipment accuracy. Equipment used to measure test conditions and test parameters shall be calibrated in accordance with MIL-STD-45662. Inspection equipment shall not introduce overall errors greater than 10 percent of the tolerances.

4.6.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions of ambient temperature, humidity, and barometric pressure:

a. Ambient temperature: 21° ± 6° C (70° ± 10° F).

b. Relative humidity: 50 ± 20 percent.

c. Barometric pressure: 30 ± 2 inches of mercury.

4.6.3 Test panels. A sufficient number of test panels shall be prepared and conditioned as specified below for performance of tests in 4.9.3 and 4.9.4.

4.6.3.1 Type I. The compound shall be catalyzed with 0.5 percent dibutyl-tin-dilaurate. Test panels measuring 6 x 6 x 0.075 inch shall be cold pressed at a temperature of 25° ± 2° C (77° ± 4° F) for 24 ± 4 hours. The test panels shall be cured for an additional period of 168 ± 4 hours at a temperature of 25° ± 2° C (77° ± 4° F) and 50 percent relative humidity. The test panels shall be conditioned at 23° ± 2° C (75° ± 4° F) and 50 percent relative humidity prior to testing.

4.6.3.2 Types II, III, and IV. The test panels shall be cured for 240 ± 8 hours at a temperature of 25° ± 2° C (77° ± 4° F) and 50 percent relative humidity. The test panels shall be conditioned 20 ± 4 hours at a temperature of 23° ± 2° C (75° ± 4° F) and 50 percent relative humidity prior to testing.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-P-14232.

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4.8 Methods of inspection.

4.8.1 Visual and mechanical. The visual and mechanical inspections shall verify compliance with the requirements indicated in 3.2, 3.3, and Section 5.

4.8.2 Uncured properties.

4.8.2.1 Specific gravity. The specific gravity of the compound shall be determined in accordance with ASTM D 792, Method A. The applicable requirements of 3.2.1a shall apply.

4.8.2.2 Nonvolatiles. The percent nonvolatiles of the compound shall be determined in accordance with MIL-S-8802. The applicable requirements of 3.2.1b shall apply.

4.8.3 Cured properties.

4.8.3.1 Hardness. The hardness of the test specimens shall be determined in accordance with ASTM D 2240. The applicable requirements of 3.2.2a shall apply.

4.8.3.2 Tensile strength. The tensile strength of the test specimens shall be determined in accordance with ASTM D 412, Die C. The applicable requirements of 3.2.2b shall apply.

4.8.3.3 Elongation. The percentage of elongation of the test specimens shall be determined in accordance with ASTM D 412, Die C. The applicable requirements of 3.2.2c shall apply.

4.8.3.4 Tear resistance. The tear resistance of the test specimens shall be determined in accordance with ASTM D 624, Die B. The applicable requirements of 3.2.2d shall apply.

4.8.3.5 Linear shrinkage. A molded specimen of compound shall be measured before and after curing. The applicable requirements of 3.2.2e shall apply.

4.8.4 Electrical properties.

4.8.4.1 Dielectric strength. The dielectric strength of the test specimens shall be determined in accordance with ASTM D 149. The applicable requirements of 3.2.4a shall apply.

4.8.4.2 Dielectric constant. The dielectric constant of the test specimens shall be determined in accordance with ASTM D 150. The applicable requirements of 3.2.4b shall apply.

4.8.4.3 Dissipation factor. The dissipation factor of the test specimens shall be determined in accordance with ASTM D 150. The applicable requirements of 3.2.4c shall apply.

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4.8.4.4 Volume resistivity. The volume resistivity of the test specimens shall be determined in accordance with ASTM D 257. The applicable requirements of 3.2.4d shall apply.

4.8.5 High temperature aging properties. Test specimens shall be subjected to the temperature specified, for the time indicated in 3.2.3a. The test specimen shall then be tested in accordance with 4.8.3.1, 4.8.3.3, and 4.8.3.4. The applicable requirements of 3.2.4 shall apply.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in contract or purchase order, preservation, packaging, and packing requirements for specified levels of protection shall be in accordance with MIL-P-14232.

5.2 Unit package size. Unit package size shall be as specified on the procurement document or purchase order.

5.3 Marking for shipment. In addition to the requirements of MIL-STD-129, unit packages shall be marked with the following as a minimum:

a. Contents (Rubber, Silicone, Type, and this specification number).

b. Quantity.

c. Vendor's name or trademark.

d. Date of manufacture.

e. Batch number.

f. Whether a catalyst is required.

g. Unit shelf life.

CAUTION:

h. Types I and II should not be used on electronic assemblies or components that are to be enclosed or sealed within 14 days, preferably 28 days after application as they evolve corrosive by-products during cure.

i. Necessary safety and toxicity information and precautions shall be listed as a warning on the unit packages.

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6. NOTES

6.1 Intended use. The compounds covered by this specification are intended to be used for adhesion, sealing, potting, vibration damping, and similar applications as specified on the applicable drawing.

CAUTION:

Types I and II should not be used on electronic assemblies or components that are to be enclosed or sealed within 14 days, preferably 28 days, after application as they evolve corrosive by-products during cure.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

b. Quantity required.

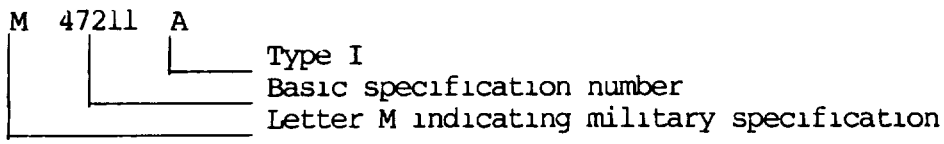
c. Type required (see 1.2).

d. First article, if required (see 3.1 and 4.4).

e. Unit package size (see 5.2).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of 10 ± 2 pounds of types I, II, III, and IV. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Part or identifying number (PIN). The PIN shall consist of the letter B, plus the basic number of this specification, followed by a letter (A for type I, B for type II, C for type III, and D for type IV) indicating type.



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6.5 Subject term (keyword) listing.

Adhesion
Compound
Sealer

6.6 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.7 Changes from previous issue. Marginal rotations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Review:
DLA - GS

Preparing Activity:
Army-MI

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-R-47211A(MI)	Rubber, Silicone, Room Temperature Curing

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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